

Grade 7 Mathematics

Մաթեմատիկա — 7-րդ դասարան

COURSE CODE	MTH-G7
PROGRAM	Middle School Mathematics · Միջին դպրոցի մաթեմատիկա
LEVEL	Grade 7 · typical age 12–13
DURATION	84–96 instructional hours (approx. 72 lessons)
PREREQUISITE	Grade 6 Mathematics or equivalent: ratio reasoning, rational number ordering, one-step equations.
LEADS TO	Grade 8 Mathematics (MTH-G8) or Algebra I (MTH-A1) on the accelerated pathway

COURSE DESCRIPTION

Դասընթացի նկարագրություն

Grade 7 completes the number system and completes proportional reasoning. Students learn to add, subtract, multiply, and divide with all rational numbers including negatives; they recognize proportional relationships, identify the constant of proportionality, and interpret it as the slope of a line through the origin. Multi-step equations and inequalities appear, along with the geometry of circles, angle relationships, and scale drawings, and probability is introduced as a measurable quantity. This is the last full year before formal algebra.

TEACHING APPROACH

Դասավանդման մոտեցումը Վարժարանում

The central Վարժարան commitment in Grade 7 is that proportionality is a *relationship*, not a procedure. A student who recognizes proportionality in a table, a graph, an equation, and a verbal description — and who can also recognize when a situation is *not* proportional — has acquired the most transferable idea in secondary mathematics. Teachers are asked to include non-proportional situations deliberately and often, because a student who has only ever seen proportional examples has learned nothing except to multiply. Operations with negatives are developed with models and with the structure of the number system, not with sign rules recited in isolation.

PLACEMENT AND READINESS INDICATORS

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The placement assessment checks each indicator below. A student missing two or more is enrolled with a remediation plan attached to the personalized curriculum.

READINESS INDICATOR	WHAT IT TELLS THE TEACHER
Solves a ratio problem with a table or diagram rather than by cross-multiplication	Predicts success in Unit 2; students who can only cross-multiply must be retaught before proportionality.
Orders negative numbers correctly and explains absolute value as distance	Prerequisite for Unit 1; a very common gap.
Solves one-step equations and checks the solution	Prerequisite for multi-step equations in Unit 3.
Computes with fractions and decimals accurately	Grade 7 combines negatives with fractions immediately; weak computation compounds.

COURSE LEARNING OUTCOMES

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On completion of this course the student will be able to:

- 1 Add, subtract, multiply, and divide rational numbers fluently, including negatives and fractions.
- 2 Apply properties of operations to compute with rational numbers and to justify each step.
- 3 Convert a rational number to a decimal and recognize that the decimal terminates or repeats.
- 4 Compute unit rates associated with ratios of fractions, including rates in different units.
- 5 Decide whether two quantities are in a proportional relationship from a table, graph, equation, or description.
- 6 Identify the constant of proportionality and interpret it in context, including as the slope of a graph.
- 7 Solve multi-step real-life problems with positive and negative rational numbers in any form.
- 8 Use variables to represent quantities and solve multi-step equations and inequalities, graphing the solution set.
- 9 Solve problems involving scale drawings, and construct triangles from given conditions.
- 10 Know and use the formulas for the area and circumference of a circle and explain the relationship between them.
- 11 Use supplementary, complementary, vertical, and adjacent angle relationships to solve for unknown angles.
- 12 Solve area, volume, and surface area problems for two- and three-dimensional objects.
- 13 Use random sampling to draw inferences about a population and compare two populations informally.
- 14 Investigate chance processes and develop, use, and evaluate probability models.

COURSE UNITS

Դասընթացի բաժիններ

Each unit below lists the lesson-level topics a teacher delivers, the objectives that define success, the observable mastery criteria used to update the student's progress profile, and the misconceptions this unit reliably produces together with the recommended teacher response.

UNIT 1 • 11-12 LESSONS

Operations with Rational Numbers

Գործողություններ ռացիոնալ թվերի հետ

TOPICS COVERED

- Adding integers on a number line and with two-color counters
- The zero pair and additive inverse
- $p + q$ as a directed distance from p
- Subtraction as adding the opposite
- The distance between two rational numbers as $|p - q|$
- Multiplying integers and why a negative times a negative is positive
- Dividing integers and the placement of the sign in a fraction
- Operations with negative fractions and decimals
- Converting rational numbers to decimals by long division
- Terminating and repeating decimals
- Applying all operations in multi-step real contexts
- Using properties of operations to compute efficiently

LEARNING OBJECTIVES

- 1 Model integer addition and subtraction on a number line and with counters.
- 2 Explain, using a pattern or the distributive property, why $(-3)(-4) = 12$.
- 3 Compute fluently with negative fractions and decimals.
- 4 Convert any rational number to a decimal and predict whether it terminates or repeats.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Computes with rational numbers at 90% accuracy across all four operations.
- Produces a justification for the negative-times-negative rule that does not rely on 'the rule says so'.
- Solves multi-step problems with mixed signs and forms in 8 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Sign rules for addition and multiplication are conflated: $-3 + -4$ answered as 12 or 7.	Separate the two families explicitly and practice them interleaved rather than in blocks. Blocked practice hides the confusion until the test.
Subtraction of a negative is left unresolved: $5 - (-3)$ answered as 2.	Model on a number line as distance and direction, and connect to the real context of removing a debt.

MISCONCEPTION	RECOMMENDED RESPONSE
-3^2 is evaluated as 9.	The exponent binds tighter than the negation. Contrast -3^2 and $(-3)^2$ deliberately and repeatedly; this error survives into calculus.

TEACHING NOTE

This unit rewards interleaved practice more than any other in the curriculum. Mixed sets of problems from the first week onward produce far better retention than one operation per lesson.

UNIT 2 • 10-11 LESSONS**Proportional Relationships**

Համեմատական կախվածություններ

TOPICS COVERED

- Unit rates with fractional quantities
- Recognizing proportionality in a table
- Recognizing proportionality in a graph: a line through the origin
- Recognizing proportionality in an equation of the form $y = kx$
- The constant of proportionality and its meaning in context
- Non-proportional linear situations and why they fail the test
- Interpreting the point $(1, k)$ on a proportional graph
- Percent increase and decrease
- Markup, discount, tax, tip, commission, and fees
- Simple interest
- Percent error
- Multi-step percent problems

LEARNING OBJECTIVES

- 1 Determine whether a relationship is proportional from any of the four representations, and justify.
- 2 Identify the constant of proportionality and state its meaning with units.
- 3 Explain why a graph passing through $(0, 5)$ is not proportional even though it is linear.
- 4 Solve multi-step percent problems including successive increases and decreases.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Classifies relationships as proportional or not, with justification, in 9 of 10 trials.
- Interprets k in context with correct units in 9 of 10 trials.
- Solves multi-step percent problems at 85% accuracy.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Any linear relationship is called proportional.	The graph must pass through the origin. Present $y = 2x$ and $y = 2x + 3$ as tables and graphs side by side in the same lesson.

MISCONCEPTION	RECOMMENDED RESPONSE
A 20% increase followed by a 20% decrease is believed to return to the original.	Compute it. The surprise is pedagogically valuable and makes the multiplicative nature of percent change memorable.
The constant of proportionality is reported as a bare number with no meaning.	Always require units and a sentence: 'k = 3.5 means the cost is \$3.50 per pound.'

TEACHING NOTE

This unit is slope, one year early and in a friendlier setting. Say so out loud in the final lesson — the connection to Grade 8 and Algebra I should be made explicit, not discovered later.

UNIT 3 • 10-11 LESSONS**Expressions, Equations, and Inequalities**

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TOPICS COVERED

- Adding, subtracting, factoring, and expanding linear expressions with rational coefficients
- Rewriting an expression to reveal how quantities are related
- Solving two-step equations of the form $px + q = r$
- Solving equations of the form $p(x + q) = r$
- Comparing arithmetic and algebraic solution paths
- Solving equations with variables on both sides
- Equations with no solution or infinitely many solutions
- Solving two-step inequalities
- Reversing the inequality symbol when multiplying or dividing by a negative
- Graphing inequality solution sets
- Interpreting an inequality solution in context
- Writing equations and inequalities from word problems

LEARNING OBJECTIVES

- 1 Solve multi-step equations with rational coefficients and check the solution.
- 2 Explain why the inequality symbol reverses when both sides are multiplied by a negative.
- 3 Write an equation or inequality modelling a real situation and interpret the solution in context.
- 4 Recognize an equation with no solution and explain what that means.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Solves multi-step equations at 90% accuracy with work shown.
- Solves and graphs inequalities correctly in 9 of 10 trials, including sign reversal.
- Translates word problems into equations correctly in 8 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
The inequality symbol is not reversed when dividing by a negative.	Demonstrate with numbers: $4 > 2$, but $-4 < -2$. Have the student verify a solution by substitution — a wrong direction fails the check immediately.
Only one side of the equation is operated on.	The balance metaphor. Require the operation to be written under both sides explicitly for the first two weeks.
The distributive property is applied to only the first term inside the parentheses.	Arrows drawn from the multiplier to every term. Return to the area model from Grade 3 if the error persists.

TEACHING NOTE

Requiring the student to check every solution by substitution is the highest-value habit in this unit. It converts equation-solving from a ritual into a verifiable claim, and it catches the majority of sign errors without teacher intervention.

UNIT 4 • 10-11 LESSONS**Geometry: Angles, Circles, and Scale**

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TOPICS COVERED

- Scale drawings and computing actual lengths and areas from a scale
- How area scales with the square of the scale factor
- Reproducing a scale drawing at a different scale
- Constructing triangles from three given measures with ruler and protractor
- Conditions determining a unique triangle, many triangles, or no triangle
- Cross-sections of right rectangular prisms and pyramids
- The circle: radius, diameter, chord, circumference
- The relationship between circumference and diameter, and the meaning of π
- Area of a circle and an informal derivation of the formula
- Supplementary, complementary, vertical, and adjacent angles
- Solving for unknown angles with an equation
- Area, volume, and surface area in composite real objects

LEARNING OBJECTIVES

- 1 Compute an actual length and an actual area from a scale drawing.
- 2 Determine from three given measures whether a unique triangle exists.
- 3 Explain where the formula $A = \pi r^2$ comes from using the rearranged-sectors argument.
- 4 Write and solve an equation for an unknown angle using a stated angle relationship.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Solves scale problems including area scaling correctly in 8 of 10 trials.
- Computes circumference and area of circles at 90% accuracy with correct units.
- Solves unknown-angle problems by writing an equation in 9 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Area is scaled by the same factor as length.	Draw a 2×3 rectangle and its double. Count squares. The factor-of-four result must be seen, not asserted.
Circumference and area formulas are interchanged.	Attach units to the formulas: circumference is a length (one dimension, one r), area is square units (two dimensions, r^2). The units make the formula self-identifying.
Vertical angles are assumed to be supplementary.	Draw and measure. Vertical angles are equal; adjacent angles on a line are supplementary. Have the student mark both on the same diagram.

TEACHING NOTE

The unknown-angle problems in this unit are the student's first taste of geometric proof: a diagram, a stated relationship, an equation, a justified conclusion. Require the reason to be written next to each step.

UNIT 5 • 7-8 LESSONS

Statistics: Sampling and Inference

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TOPICS COVERED

- Populations and samples
- Why a sample must be random to be representative
- Identifying sources of bias in a sampling method
- Estimating a population characteristic from a sample
- Variation among samples from the same population
- Simulating repeated sampling to see the variation
- Comparing two populations using centers and spreads
- Expressing a difference in centers as a multiple of the variability
- Drawing an informal comparative inference and stating its limits

LEARNING OBJECTIVES

- 1 Explain why random selection is required for a sample to support an inference.
- 2 Identify the bias in a described sampling method and propose a better one.
- 3 Estimate a population mean or proportion from a sample and describe the uncertainty.
- 4 Compare two data distributions and state a defensible informal conclusion.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Identifies sampling bias correctly in 8 of 10 described scenarios.
- Produces a reasonable population estimate with a statement of uncertainty.
- Writes a comparative inference that references both center and spread.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
A large sample is assumed to be representative regardless of how it was selected.	Size does not fix bias. Use a memorable historical example of a large biased sample and let the student diagnose it.
Two distributions are compared by their means alone.	Require both center and spread in every comparison statement; overlapping distributions with different means may not support a conclusion at all.

TEACHING NOTE

This unit is genuinely the foundation of AP Statistics. The idea that a conclusion about a population carries uncertainty is new and difficult; give it real time rather than treating it as an end-of-year appendix.

UNIT 6 • 8-9 LESSONS**Probability**

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TOPICS COVERED

- Probability as a number between 0 and 1 and the meaning of the endpoints
- Likely, unlikely, and equally likely events
- Theoretical probability from equally likely outcomes
- Experimental probability from collected data
- The relationship between the two as trials increase
- Developing and evaluating a uniform probability model
- Developing a model from observed frequencies when outcomes are not equally likely
- Compound events and sample spaces
- Organized lists, tables, and tree diagrams
- Probability of a compound event
- Simulation to estimate a compound probability

LEARNING OBJECTIVES

- 1 Compute theoretical probability from a sample space and experimental probability from data.
- 2 Explain why experimental probability approaches theoretical probability as trials increase.
- 3 Construct a sample space for a compound event with a table or tree diagram.
- 4 Design and run a simulation to estimate a probability that is hard to compute directly.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Computes both probability types correctly in 9 of 10 trials.
- Constructs a correct sample space for a two-stage compound event in 4 of 5 trials.
- Designs a valid simulation for a stated problem.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
The gambler's fallacy: after four heads, tails is believed more likely.	Coins have no memory. Simulate 200 flips and examine the runs; the data is more persuasive than the explanation.
An outcome is called 'impossible' when it merely has low probability.	Reserve probability 0 for genuinely impossible events. Precision here matters for all later work.
Sample spaces for compound events are enumerated unsystematically and come out incomplete.	Require a table or tree every time until the student's lists are reliably exhaustive.

TEACHING NOTE

Simulation is the most valuable technique in this unit and the one most often skipped. A student who can design a simulation understands probability structurally, not just computationally.

ASSESSMENT AND MASTERY POLICY

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ASSESSMENT	WHEN	WHAT IT MEASURES
Placement Assessment	Before enrollment	Grade 6 ratio reasoning and rational-number ordering; identifies whether cross-multiplication has replaced understanding.
Unit Check	End of each unit · 40 min	Unit mastery criteria plus two items requiring written justification.
Fluency Probe	Every 3rd lesson · 5 min	Rational-number computation, interleaved across all four operations.
Proportionality Diagnostic	After Unit 2	Includes deliberately non-proportional situations; the key predictor of Algebra I readiness.
Mid-Year Assessment	After Unit 3	Cumulative Units 1–3; weighted toward rational numbers and equations.
End-of-Course Assessment	After Unit 6	All course outcomes; determines whether the student proceeds to Grade 8 or directly to Algebra I.

Վարժարան reports mastery, not grades. Each topic in the student's profile is marked **Mastered**, **Developing**, or **Needs Review** against the criteria stated in each unit above. A topic is marked Mastered only when the criterion is met on two separate occasions at least one week apart — this prevents short-term recall from being recorded as durable learning. Topics marked Needs Review are automatically re-entered into homework and into the opening minutes of subsequent lessons until they are re-mastered.

PACING OPTIONS

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TRACK	CADENCE	EXPECTED COMPLETION
Standard	1 lesson/week, 60 min	Full course in approximately 50–55 weeks.
Accelerated	2 lessons/week	Full course in approximately 28–32 weeks.
Compacted 7/8	2 lessons/week	Grade 7 and Grade 8 compressed for students targeting Algebra I in Grade 8 with a strong Grade 6 profile.
Intensive / Catch-Up	2–3 lessons/week	22–26 weeks with diagnostic-driven ordering.
Summer Program	3 lessons/week, 6 weeks	Rational number operations intensive (Unit 1) or proportional reasoning intensive (Unit 2).

Pacing is assigned after the placement assessment and reviewed at the mid-year assessment. A student who is meeting mastery criteria early may be moved to the accelerated track; a student who is not may be moved to intensive without any change in the course content itself.

HOMEWORK AND INDEPENDENT PRACTICE

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Forty minutes per lesson: seven to ten problems. Practice is deliberately interleaved rather than blocked — an assignment on multiplying negatives will also contain addition and subtraction problems, because the confusion between the sign rules only surfaces when the operations are mixed. Every assignment contains at least one problem asking the student to explain why a stated method works or why a given answer is wrong. From Unit 3 onward, solutions to equations must be checked by substitution and the check must be shown.

PROGRESS MILESTONES REPORTED TO PARENTS

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These are the achievements the parent portal announces as the student reaches them. They replace attendance counts as the primary measure of progress.

- ◆ Adds, subtracts, multiplies, and divides all rational numbers fluently.
- ◆ Justifies the sign rules rather than reciting them.
- ◆ Recognizes proportional relationships in tables, graphs, equations, and descriptions.
- ◆ Identifies and interprets the constant of proportionality with units.
- ◆ Solves multi-step percent problems including markup, discount, and interest.
- ◆ Solves multi-step equations and inequalities with rational coefficients.
- ◆ Solves scale-drawing problems and understands how area scales.
- ◆ Knows and can justify the circumference and area formulas for a circle.
- ◆ Solves unknown-angle problems by writing and solving equations.
- ◆ Draws informal inferences from random samples and states their limits.
- ◆ Builds and evaluates probability models and designs simulations.
- ◆ Completed Grade 7 Mathematics — ready for Grade 8 or Algebra I.

MATERIALS AND RESOURCES

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- Two-color counters and an extended number line for integer operations.
- Compass, protractor, and ruler for the geometry unit (physical set at home required).
- Circular objects and string for the π investigation.
- Random-number generator and simulation tool for the statistics and probability units.
- Վարժարան Grade 7 problem bank — approximately 1,400 items with interleaved practice sets built in.

Վարժարան · Varzharan — Exceptional Armenian educators. American academic standards. Personalized education. | This syllabus defines the standard course. Every enrolled student receives a personalized version of it, sequenced from their placement assessment and revised as their progress profile changes.